

Neural quantum states for the simulation quantum many-body systems

Zakari Denis

CECAM school: Machine Learning–Assisted Sampling
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$$\psi(\mathbf{X}) = f_{\theta} \left(\begin{array}{c} \text{Diagram of a graph with 5 nodes and green edges. Two nodes at the top are labeled } x_i \text{ and } x_j \text{ with red wavy arrows pointing to them. A label } V_{ij} \text{ is placed above the edge connecting } x_i \text{ and } x_j. \end{array} \right)$$

Why neural quantum states?

Central task in computational quantum physics

Determine the properties of a system from its Hamiltonian. Formally, solve:

$$\hat{H} |\psi_n\rangle = E_n |\psi_n\rangle$$

$$|\psi\rangle = \sum_{x \in \mathcal{X}} \psi(x) |x\rangle$$

Exponentially
many amplitudes

$$\mathcal{X} = \{\uparrow, \downarrow\}^N \quad \dim \mathcal{H} = 2^N$$

$$N = 20 \longrightarrow 16 \text{ MB}$$

$$N = 50 \longrightarrow 16 \text{ PB}$$

$$N = 100 \longrightarrow 2 \times 10^{31} \text{ bytes}$$

Exact diagonalisation

Tensor networks

Quantum Monte Carlo

Neural quantum states



Exact

Robust and principled

Large system sizes

Flexible

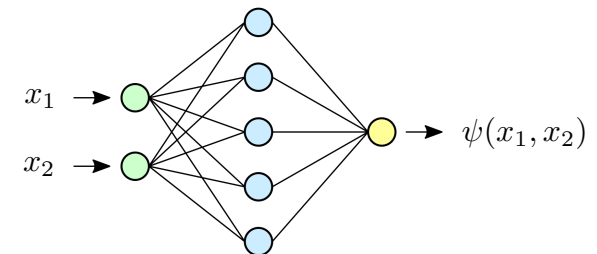


Small system sizes

Limited entanglement

Sign problem

Unclear



Variational approach

Throughout this course, we focus on the determination of the ground state

$$\hat{H} |\psi_0\rangle = E_0 |\psi_0\rangle$$

Lowest energy

For any $|\psi\rangle \in \mathcal{H}$

$$E[\psi] = \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} = \frac{\sum_n |c_n|^2 E_n}{\sum_n |c_n|^2} \geq E_0$$

$$|\psi\rangle = \sum_n c_n |\psi_n\rangle$$

Variational principle

$$|\psi_0\rangle = \arg \min_{\psi \in \mathcal{H}} E[\psi]$$

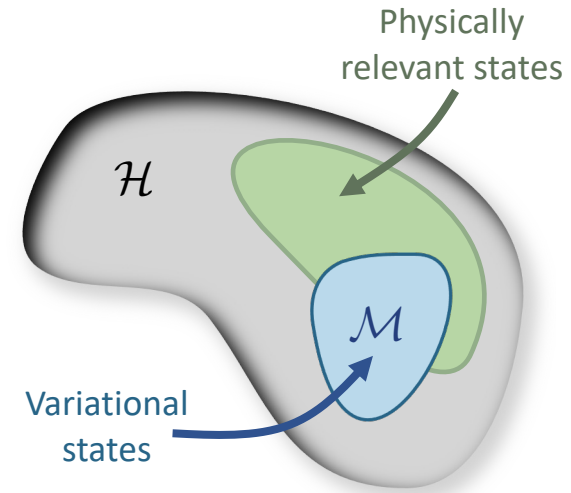
Ground-state search becomes an optimisation problem!

Variational manifold

$$|\psi\rangle = \sum_x \psi(x) |x\rangle$$

$$|\psi_\theta\rangle = \sum_x \psi_\theta(x) |x\rangle$$

$$\mathcal{M} := \{|\psi_\theta\rangle \mid \theta \in \mathbb{R}^p\}$$



$$|\psi_\theta\rangle_{\text{GS}} = \arg \min_{\psi \in \mathcal{M}} E[\psi]$$

$$\theta_{\text{GS}} = \arg \min_{\theta \in \mathbb{R}^p} E_\theta$$



Optimisation becomes tractable

Variational approach

Optimisation: naïve ML approach

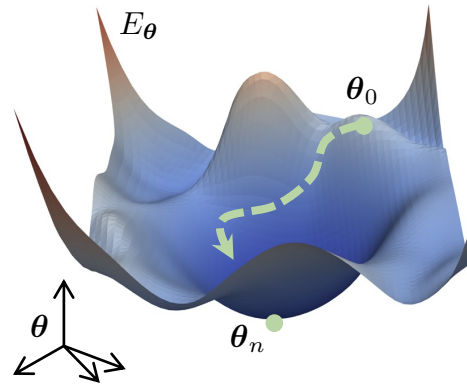
Gradient-descent algorithm:

$\theta_0 \leftarrow$ random initial values

$\theta_1 \leftarrow \theta_0 - \eta \partial_{\theta} E_{\theta} |_{\theta=\theta_0}$

Learning rate $\dots$

$\theta_n \leftarrow \theta_{n-1} - \eta \partial_{\theta} E_{\theta} |_{\theta=\theta_{n-1}}$



Central object: variational energy

$$E_{\theta} := \frac{\langle \psi_{\theta} | \hat{H} | \psi_{\theta} \rangle}{\langle \psi_{\theta} | \psi_{\theta} \rangle} \geq E_0$$

Analog to a loss in machine learning

Evaluation of the variational energy

$$E_{\theta} = \frac{\sum_{xy} \psi_{\theta}(x) H_{xy} \psi_{\theta}(y)}{\sum_x |\psi_{\theta}(x)|^2}$$

with $H_{xy} := \langle x | \hat{H} | y \rangle$

→ Cost is still exponential though (?)

$$E_{\theta} = \sum_x \underbrace{\left(\frac{|\psi_{\theta}(x)|^2}{\sum_y |\psi_{\theta}(y)|^2} \right)}_{=: p_{\theta}(x) \text{ Born probability}} \underbrace{\sum_y H_{xy} \frac{\psi_{\theta}(y)}{\psi_{\theta}(x)}}_{=: E_{\text{loc}}(x) \text{ local energy}}$$

$$= \sum_x p_{\theta}(x) E_{\text{loc}}(x)$$

Just a classical average

Variational approach

Evaluation of the variational energy

$$E_{\theta} = \sum_x p_{\theta}(x) E_{\text{loc}}(x)$$

1 Exploit the Hamiltonian's sparsity

$$\begin{aligned} E_{\text{loc}}(x) &= \sum_y H_{xy} \frac{\psi_{\theta}(y)}{\psi_{\theta}(x)} \\ &= \sum_{y \in \mathcal{C}_H(x)} H_{xy} \frac{\psi_{\theta}(y)}{\psi_{\theta}(x)} \end{aligned}$$

Connected elements:

$$\mathcal{C}_H(x) = \{y \mid \langle x | \hat{H} | y \rangle \neq 0\}$$

Matrix elements:

$$H_{xy} := \langle x | \hat{H} | y \rangle$$

polynomially many

2

Evaluation of the variational energy

$$\hat{E}_{\theta} = \frac{1}{N_s} \sum_{i=1}^{N_s} E_{\text{loc}}(x^{(i)}), \quad x^{(i)} \sim p_{\theta}$$

Law of large numbers: $\hat{E}_{\theta} \xrightarrow[N_s \rightarrow +\infty]{} E_{\theta}$

Unbiased: $\mathbb{E}[\hat{E}_{\theta}] = E_{\theta}$

Statistical error: $\Delta E_{\theta} = \sqrt{\text{Var}[E_{\text{loc}}]/N_s}$

Zero-variance property

$$\hat{H} |\psi_{\theta}\rangle = E |\psi_{\theta}\rangle \Rightarrow E_{\text{loc}}(x) = \frac{\langle x | \hat{H} | \psi_{\theta} \rangle}{\langle x | \psi_{\theta} \rangle} = E$$

As we approach the ground state, the Monte Carlo estimation becomes increasingly accurate

Variational approach

Same idea for the gradients

$$\partial_{\theta_k} E_{\theta} = 2\text{Re} \left[\overline{O_k^* E_{\text{loc}}} - \overline{O_k^*} \overline{E_{\text{loc}}} \right]$$

average over samples

Normalisation
contribution

Logarithmic derivatives:

$$O_k(x) := \partial_{\theta_k} \log \psi_{\theta}(x)$$

Markov chain Monte Carlo (MCMC)

Goal

Sample from $p_{\theta}(x) = \frac{1}{Z} |\psi_{\theta}(x)|^2$

Without ever computing the normalisation constant

$$Z = \sum_x |\psi_{\theta}(x)|^2$$

We construct a Markov chain

$$x^{(0)} \rightarrow x^{(1)} \rightarrow x^{(2)} \rightarrow \dots$$

Such that the stationary distribution relaxes to p_{θ}

Ergodicity

&

Stationarity

MCMC algorithm

Starting from a configuration x :

(i) Propose a new configuration

$$x' \sim T(x \rightarrow x')$$

(ii) Accept with probability

$$A(x \rightarrow x') = \min \left(1, \frac{p(x')T(x' \rightarrow x)}{p(x)T(x \rightarrow x')} \right)$$

(iii) Otherwise remain in x

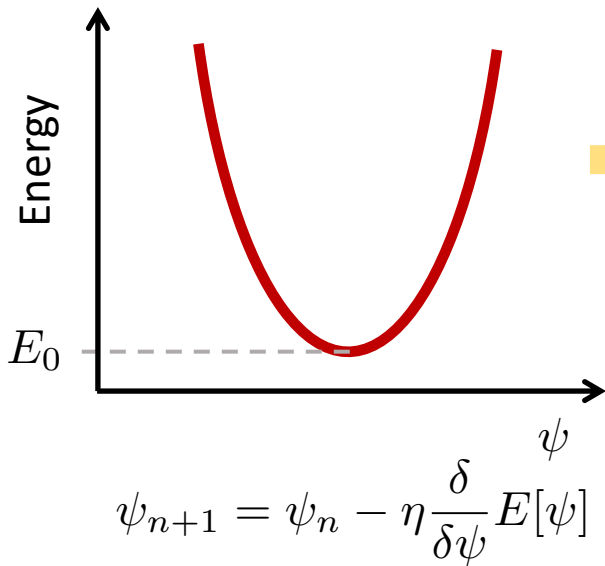
For symmetric proposals

$$A(x \rightarrow x') = \min \left(1, \frac{|\psi_{\theta}(x')|^2}{|\psi_{\theta}(x)|^2} \right)$$

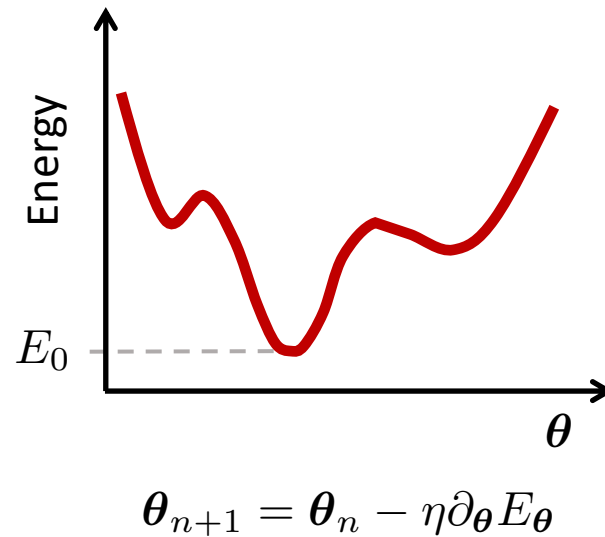
normalisations
cancel out

Optimising the variational wavefunction

$$|\psi\rangle_{\text{GS}} = \arg \min_{\psi \in \mathcal{H}} E[\psi]$$



$$\theta_{\text{GS}} = \arg \min_{\theta \in \mathbb{R}^p} E_\theta$$



Tremendous simplification, but at what cost?

→ The geometry of the Hilbert space

Distance between two
parameter sets



Distance between two
wavefunction

Can we do better?

$$\partial_\tau |\psi(\tau)\rangle = -\hat{H} |\psi(\tau)\rangle$$

$\Leftrightarrow$

$$|\psi(\tau)\rangle = e^{-\hat{H}\tau} |\psi(0)\rangle$$

$$= \sum_n c_n e^{-\tau \hat{H}} |\psi_n\rangle$$

$$|\psi(0)\rangle = \sum_n c_n |\psi_n\rangle$$

$$\Rightarrow \frac{|\psi(\tau)\rangle}{|\psi(\tau)|} = \tilde{c}_n |\psi_0\rangle + \sum_{n>0} \tilde{c}_n e^{-\tau(E_n - E_0)} |\psi_n\rangle$$

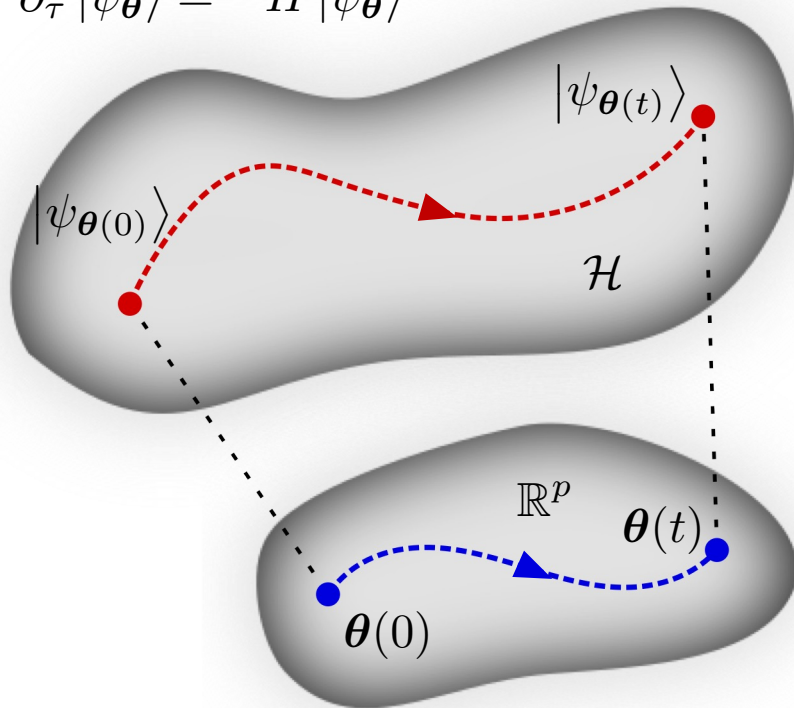
Imag. dynamics

Exponential convergence at a rate

$$1/(E_1 - E_0)$$

Optimising the variational wavefunction

$$\partial_\tau |\psi_\theta\rangle = -\hat{H} |\psi_\theta\rangle$$



?

Variational principle

$$\dot{\theta} = \arg \min_{\dot{\eta}} d_{\text{FS}} \left(e^{-\hat{H}\delta\tau} |\psi_\theta\rangle, |\psi_{\theta+\delta\tau\dot{\eta}}\rangle \right)$$



Stochastic-reconfiguration (SR) equation

$$\dot{\theta} = \mathbf{S}^{-1} \mathbf{F}$$

$$S_{kk'} = \overline{(O_k^* - \overline{O_k^*})(O_{k'} - \overline{O_{k'}})}$$

Quantum geometric tensor

$$F_k = \overline{(O_k^* - \overline{O_k^*})(E_{\text{loc}} - \overline{E_{\text{loc}}})}$$

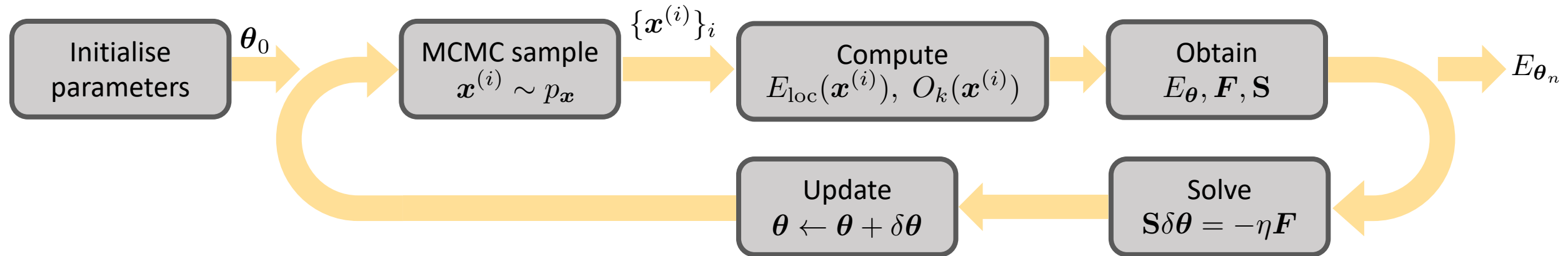
Forces $\sim$ energy gradient

SR rescales the gradient according to the physical effect of each parameter on the wavefunction.

$$\theta_{n+1} = \theta_n - \eta \partial_\theta E_\theta \quad \longrightarrow \quad \theta_{n+1} = \theta_n - \eta \mathbf{S}^{-1} \partial_\theta E_\theta$$

Optimising the variational wavefunction

The VMC algorithm



Neural quantum states

G. Carleo and M. Troyer, Science 335(6325), 602-606 (2017)

Fundamental limit

$$|\psi_{\theta}\rangle_{\text{GS}} = \arg \min_{\psi \in \mathcal{M}} E[\psi], \quad \mathcal{M} := \{|\psi_{\theta}\rangle \mid \theta \in \mathbb{R}^p\}$$

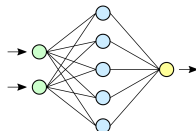
The quality of the approximation is ultimately determined by the choice of variational manifold.

$$|\psi\rangle = \sum_x \psi(x) |x\rangle$$

$$\psi : \mathcal{X} \longrightarrow \mathbb{C}$$

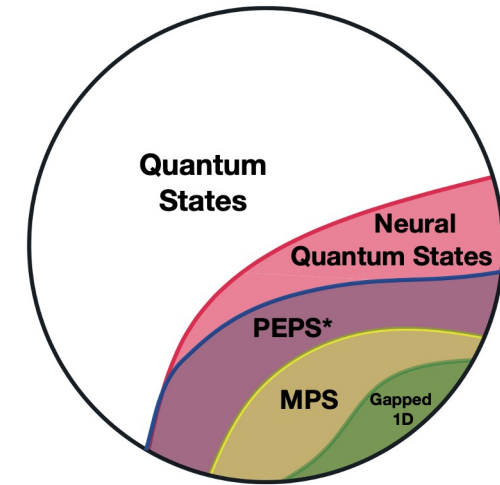
Just a high-dimensional function

Can we learn it with a flexible family of functions?



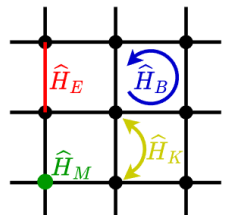
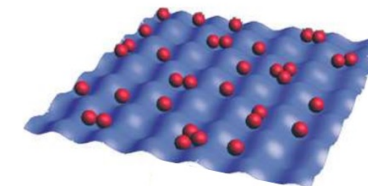
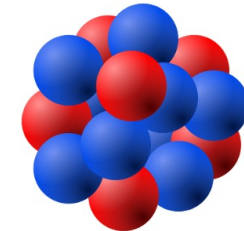
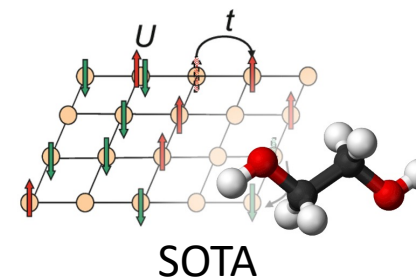
Universal approximator (Cybenko, Hornik, etc.)

O. Sharir, A. Shashua, and G. Carleo, Phys. Rev. B **106**, 205136 (2022)



No direct limitations in terms of entanglement!

ZD, Sinibaldi, and Carleo, PRL **134**, 079701 (2025)
and references therein



Neural quantum states

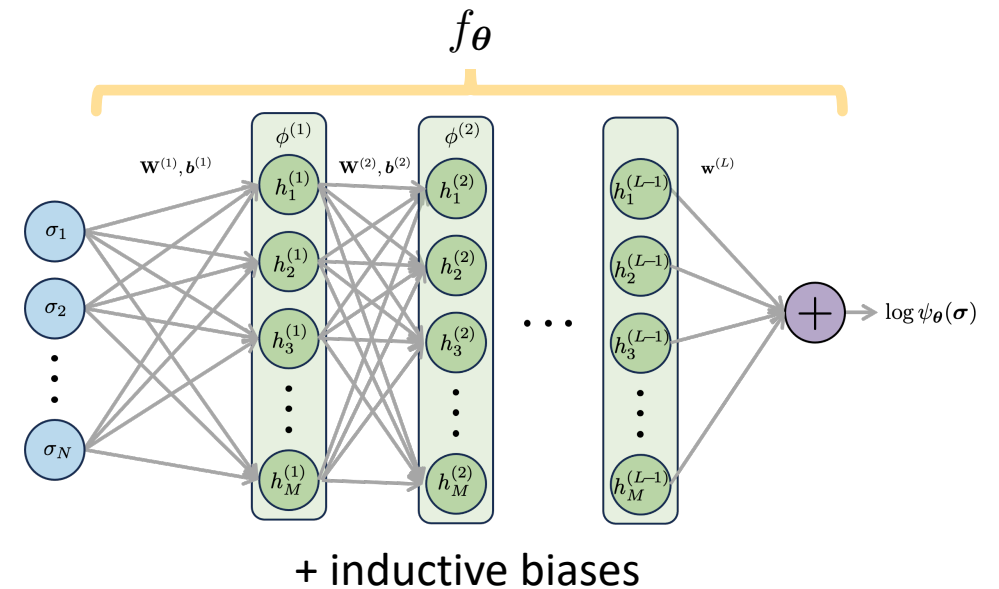
G. Carleo and M. Troyer, Science 335(6325), 602-606 (2017)

Neural quantum state

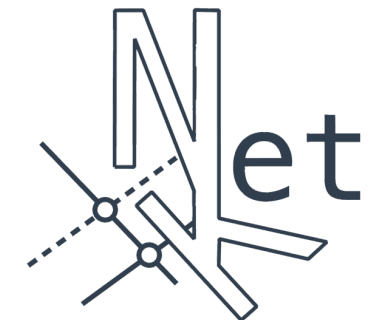
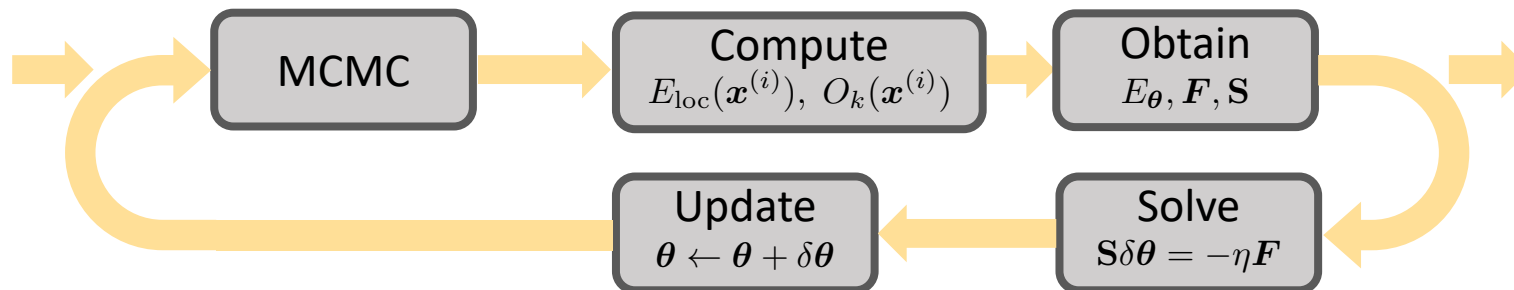
$\log \psi_{\theta}(\mathbf{x}) = f_{\theta}(\mathbf{x})$ ← Work in log space

e.g. L-layer perceptron $f_{\theta}(\mathbf{x}) = \mathbf{w}^{(L)} \cdot \mathbf{h}^{(L-1)}$.

$$\mathbf{h}^{(\ell+1)} = \phi^{(\ell)} \left(W^{(\ell)} \mathbf{h}^{(\ell)} + \mathbf{b}^{(\ell)} \right), \quad \mathbf{h}^{(0)} = \mathbf{x},$$



All the VMC machinery remains the same:



`pip install --upgrade netket`

F. Vicentini *et al.*, SciPost Phys. Codebases, 7 (2022)

Any questions?